# Md Tanvirul Alam

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Google Scholar | GitHub | LinkedIn

ML research engineer developing learning and tool-using agent systems across language, vision, and structured domains. Work spans model adaptation, controlled experimentation, and robust evaluation across research and industry ML systems.

## Technical Skills

**Languages / Frameworks**: Python, C, C++, Java; PyTorch, Hugging Face Transformers, veRL, TensorFlow, scikit-learn
**Model Adaptation**: SFT, QLoRA, RLHF/RLAIF, RLVR/GRPO, distillation, RAG, hybrid sparse–dense retrieval
**Research Methods**: Experimental design, ablation studies, benchmark development, error analysis, robustness and distribution-shift analysis
**Agent Systems / Evaluation**: Tool-using agents, structured tool interfaces, procedural environments, hidden evaluators, budget-aware planning, auditable episode traces
**Data / Systems**: Procedural and synthetic data generation, distributed multi-GPU training, structured data pipelines, real-time and edge inference

## Experience

**Graduate Research Assistant** 2021 – 2026
*Rochester Institute of Technology* *Rochester, NY*
- Designed controlled training and evaluation studies across LLM/VLM reasoning, structured extraction, robustness, and distribution shift, including dynamic assessment of 12 LLMs.
- Built reusable PyTorch/veRL pipelines for procedural dataset construction, distributed training, deterministic reward computation, ablation studies, and cross-benchmark analysis.
- Developed model-adaptation methods using SFT, QLoRA, RLVR/GRPO, distillation, retrieval augmentation, and synthetic-data-driven training.
- Released CTIBench and AthenaBench, later used in official model evaluations by Google, Cisco Foundation AI, Trend Micro AI Lab, and the Qwen team.
- Built and evaluated tool-using agents for resource-bounded scientific planning and multi-step defensive threat hunting using structured tools, procedural environments, hidden evaluators, and auditable traces.

**Research Intern** Jul. 2025 – Jan. 2026
*Athena Security Group* *Remote*
- Built structured training and evaluation tasks from authoritative technical sources, with explicit answer formats, deterministic checks, and repeatable experiment pipelines.
- Compared SFT, GRPO/RLVR, and data-design interventions to isolate how task difficulty, reward sparsity, and feedback quality affected model outcomes.

**Senior Software Engineer** May 2018 – Jun. 2021
*BJIT Limited* *Dhaka, Bangladesh*
- Delivered active-speaker framing for a real-time 360-degree meeting assistant; served as team lead across 5+ client-facing ML projects spanning vision, audio, NLP, and safety.
- Built real-time facial-attribute and active-speaker systems, covering custom data capture, model training, evaluation, and application integration.

**Software Engineer** Dec. 2016 – Apr. 2018
*Semion Limited* *Dhaka, Bangladesh*
- Built SemInfer, a dependency-free C++ inference engine for CNN and object-detection models used by internal applications; matched TensorFlow classification scores to six decimal places across ported models.
- Built the doctor-side JavaFX application for DICOM chest X-rays, integrating model predictions with image annotation and radiologist-facing report preparation.

## Selected Projects

**Trace: A Taxonomy-Guided Environment for Multidomain Visual Reasoning** Code (Focus: VLMs, RLVR, procedural data)
- Built a visual-reasoning environment with 1,000 tasks, 277 scene grammars, and 11 visual domains using a deterministic generator-renderer-verifier pipeline.
- Ran full-parameter GRPO on Qwen2.5-VL-3B/7B using 64K prompts and 512K sampled responses.
- Improved the macro-average across 24 external benchmarks by 3.51 and 4.06 percentage points at 3B and 7B, respectively.

**Minerva: RLVR Post-Training for Structured Domain LLMs** Paper (Focus: LLMs, RLVR, reward sparsity)
- Built a 32K-example post-training suite with 16 structured reasoning tasks and evaluated four Llama and Qwen backbones across 12 benchmarks.
- Developed MinervaRL to generate, filter, and distill reasoning traces for prompts where sparse rewards limited standard RLVR training.
- Improved mean score by 15.8 percentage points over base models and 4.3 points over GRPO.

## Selected Projects (continued)

**CTIBench / AthenaBench: Dynamic Evaluation Infrastructure for Domain LLMs** CTIBench Code (Focus: LLM evaluation, structured reasoning)

- Built five structured task families with task-specific answer formats and scoring logic for knowledge QA, vulnerability analysis, technique extraction, and threat attribution.
- Extended the benchmark into an 8.1K-sample evaluation pipeline from live technical sources and evaluated 12 LLMs on continuously updated tasks.

**SPHINX: Synthetic Visual Reasoning Environment** Code (Focus: VLMs, synthetic data, evaluation)

- Built a 25-task synthetic visual-reasoning environment spanning spatial transformation, counting, topology, chart interpretation, symmetry, and sequence reasoning.
- Evaluated 19 LVLMs plus humans across IID and distribution-shifted variants to characterize visual abstraction and reasoning failures.
- Trained Qwen3-VL-4B/8B on 32K samples, improving accuracy by 12.9 points on IID tasks, 6.4 points on OOD tasks, and 2.6 points on average across eight external benchmarks.

**VLM-Fix / VLMBias: Controlled Analysis of VLM Behavior** (Focus: VLM behavior, robustness, post-training)

- Built controlled evaluations that separate visual-perception failures from failures to apply alternative semantic rules and tested interventions across 14 VLMs.
- Improved average accuracy from 11.6% to 20.7% while reducing bias from 76.7% to 58.9% using neutral and defamiliarized rule mappings.
- Analyzed post-training transfer and late-layer representations, finding rule-dependent transfer and partially recovering degraded inverse-rule performance through activation steering.

**Actionable CTI with LLMs and Knowledge Graphs** (Focus: QLoRA, extraction, knowledge graphs)

- Fine-tuned Llama 2-7B Chat with QLoRA for structured subject-relation-object extraction from unstructured technical reports, reaching 0.567 ROUGE-L.
- Scaled the extraction pipeline across 12K reports and 80K paragraphs, refining 77,307 generated triples into 41,525 ontology-filtered triples.

**Adapting LLMs to Emerging Cybersecurity with Hybrid RAG** (Focus: RAG, FAISS, BM25, hybrid retrieval)

- Developed and evaluated a RAG pipeline for Llama-3-8B-Instruct combining FAISS dense retrieval, BM25 sparse retrieval, and CVE-aware matching over official CVE and CWE sources.
- Compared baseline and hybrid retrieval, embedding models, and decoding temperatures; hybrid retrieval with CVE matching improved fully automated KCV accuracy from 57.6% to 72.7%, while hybrid retrieval improved CWET accuracy from 86.4% to 92.2%.

**Disaster-Response Image Classification** (Focus: model training, multi-task learning, robustness)

- Trained and evaluated image-classification models across real-world disaster-response tasks and datasets containing up to 71K social-media images.
- Compared ten vision architectures across four tasks and evaluated data augmentation, semi-supervised training, and multi-task learning strategies.

## Education

**Rochester Institute of Technology** (2026), Ph.D. in Computing and Information Sciences
**Bangladesh University of Engineering & Technology** (2011 – 2016), B.Sc. in Electrical and Electronic Engineering

## Selected Publications

**Minerva: Reinforcement Learning with Verifiable Rewards for Cyber Threat Intelligence LLMs** – TMLR 2026.
**SPHINX: A Synthetic Environment for Visual Perception and Reasoning** – CVPR 2026.
**CTIBench: A Benchmark for Evaluating LLMs in Cyber Threat Intelligence** – NeurIPS 2024 Spotlight.
**Trace: A Taxonomy-Guided Environment for Multidomain Visual Reasoning** – arXiv, 2026.
**Beyond Perception Errors: Semantic Fixation in Large Vision-Language Models** – arXiv 2026.
**Actionable Cyber Threat Intelligence using Knowledge Graphs and Large Language Models** – IEEE EuroS&PW 2024.
**Adapting Large Language Models to Emerging Cybersecurity using Retrieval Augmented Generation** – arXiv:2510.27080, 2025.
**Robust Training of Social Media Image Classification Models** – IEEE TCSS 2024.
**Limits of Generalization in RLVR: Two Case Studies in Mathematical Reasoning** – NeurIPS 2025 Math-AI Workshop.